

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0504
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COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I Geferlin Boffy R (192571080) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS

Q1 Schema

A **schema** is the overall structure or blueprint of a database. It defines how data is organized, including tables, attributes, relationships, and constraints. A schema changes very rarely.

Example

A Student database may contain:

Student(Student_ID, Name, Department, Age)

Here, the table structure is the **schema**.

Instance

An **instance** is the actual data stored in the database at a particular point in time. Unlike the schema, the instance changes whenever records are inserted, updated, or deleted.

Example

Student_ID Name Department Age

101 Rahul CSE 20

102 Priya ECE 19

This table is an **instance** of the Student schema.

Difference between Schema and Instance

Schema	Instance
Blueprint of the database	Actual data stored
Changes rarely	Changes frequently
Defines structure	Represents current data
Designed by DBA	Updated by users/applications

Types of Schema

1. Logical Schema

- Describes the logical structure of the database.
- Specifies tables, attributes, relationships, and constraints.
- Independent of storage details.

Example:

Student(Student_ID, Name, Department)

2. Physical Schema

- Describes how data is physically stored.
- Includes storage blocks, indexes, file organization, hashing, etc.
- Managed by the DBMS.

Example

Student table stored in:

- B+ Tree Index
- 4 KB disk pages
- Heap file organization

3. View Schema (External Schema)

- Represents only the part of the database visible to a particular user.
- Improves security and simplifies access.

Example

Teacher View:

| Student_ID | Name | Marks |

Student View:

| Student_ID | Name |

Different users see different views of the same database.

Comparison

Logical Schema	Physical Schema	View Schema
Defines database structure	Defines storage details	Defines user view
Used by designers	Used by DBMS	Used by end users

Logical Schema	Physical Schema	View Schema
Independent of hardware	Depends on storage	Shows partial data
Moderate changes	Rare changes	Frequently customized

Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Entities

Customer

Attributes:

- Customer_ID (Primary Key)
- Name
- Address
- Phone
- Email

Account

Attributes:

- Account_No (Primary Key)
- Account_Type
- Balance
- Opening_Date

Branch

Attributes:

- Branch_ID (Primary Key)
- Branch_Name
- Location

Loan

Attributes:

- Loan_ID (Primary Key)
- Loan_Amount
- Interest_Rate

Employee

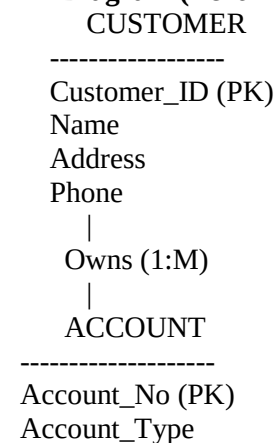
Attributes:

- Employee_ID (Primary Key)
- Name
- Designation
- Salary

Relationships

1. Customer **owns** Account (One customer can own many accounts)
2. Branch **maintains** Account (One branch maintains many accounts)
3. Customer **takes** Loan (One customer may take multiple loans)
4. Branch **issues** Loan
5. Employee **works in** Branch

ER Diagram (Text Representation)



Balance
Opening_Date
|
Maintained By (M:1)
|
BRANCH

Branch_ID (PK)
Branch_Name
Location
|
Issues (1:M)
|
LOAN

Loan_ID (PK)
Loan_Amount
Interest_Rate
^
|
Takes (1:M)
|
CUSTOMER

EMPLOYEE

Employee_ID (PK)
Name
Designation
Salary
|
Works In (M:1)
|
BRANCH

Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

Extended ER (EER) Model

The **Extended Entity-Relationship (EER) Model** is an advanced version of the ER model. It introduces inheritance, subclasses, superclass relationships, and abstraction concepts to represent complex real-world applications.

Advantages

- Models real-world situations accurately.
- Supports inheritance.
- Reduces redundancy.
- Improves database design.

Generalization

Generalization combines multiple lower-level entities into one higher-level entity based on common attributes.

Example

Savings Account
Current Account
↓
Account

Common attributes like Account Number and Balance are placed in the superclass.

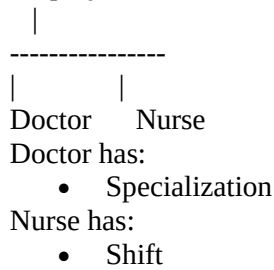
Specialization

Specialization divides one higher-level entity into multiple lower-level entities based on distinguishing

characteristics.

Example

Employee



Aggregation

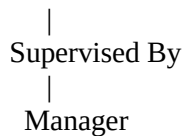
Aggregation represents a relationship between relationships.

Example

A Manager supervises an Employee working on a Project.

Instead of connecting separately, the relationship "Works_On" is treated as a higher-level entity, allowing another relationship "Supervises."

Employee ---- Works_On ---- Project



Comparison

Concept	Meaning
Generalization	Bottom-up approach; combines entities
Specialization	Top-down approach; divides entities
Aggregation	Treats a relationship as an entity

Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

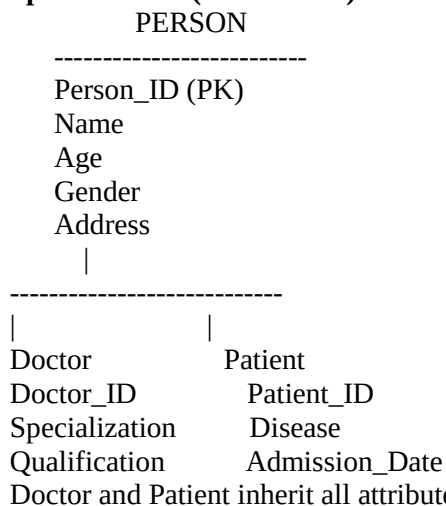
Superclass

Person

Attributes:

- Person_ID (PK)
- Name
- Age
- Gender
- Address

Specialization (Inheritance)



Other Entities

Department

- Department_ID (PK)
- Department_Name

Appointment

- Appointment_ID (PK)
- Date
- Time

Room

- Room_No (PK)
- Type

Weak Entity

Medical_Record

Attributes:

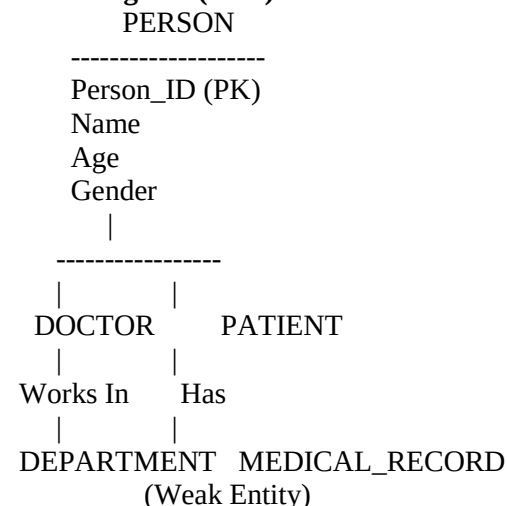
- Record_No (Partial Key)
- Diagnosis
- Prescription

Medical_Record depends on Patient and cannot exist independently.

Relationship:

Patient → Medical_Record (1:M)

EER Diagram (Text)



PATIENT ---- Books ---- APPOINTMENT ---- DOCTOR

PATIENT ---- Admitted To ---- ROOM

Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Database Administrator (DBA)

A **Database Administrator (DBA)** is responsible for managing, maintaining, securing, and ensuring the efficient operation of databases within an organization.

Responsibilities of a DBA

1. Database Installation

- Install DBMS software.
- Configure database servers.

2. Database Design

- Create databases, tables, indexes, and schemas.
- Define constraints and relationships.

3. Security Management

- Create user accounts.
- Assign permissions.

- Prevent unauthorized access.

4. Backup and Recovery

- Perform regular backups.
- Restore databases after failures.

5. Performance Tuning

- Optimize SQL queries.
- Create indexes.
- Improve response time.

6. Data Integrity

- Enforce constraints.
- Maintain data consistency and accuracy.

7. Monitoring

- Monitor storage usage.
- Detect errors and system failures.
- Ensure database availability.

8. User Support

- Help users resolve database issues.
- Provide technical guidance.

Importance of a DBA

- Ensures data security.
- Maintains database availability.
- Prevents data loss.
- Improves performance.
- Supports organizational decision-making.
- Ensures reliable and efficient database operations.